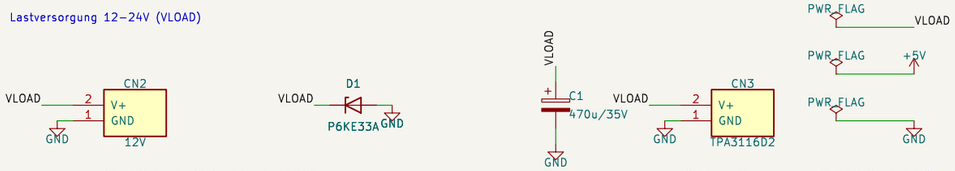
CRSF Multiswitch + Sound, ESP32-S3-DevKitC-1: 8 Kanaele (ULN2803A), 3 UART, I2S (PCM5102A oder MAX98357A), TPA3116D2, microSD – Entwurf v0.1

Last max. 24V / 500mA je Kanal. ESP32 aus +5V von CN1. TPA3116D2-Modul direkt aus VLOAD (CN3). Audio: nur EIN Modul bestuecken (PCM5102A + TPA3116D2 ODER MAX98357A).

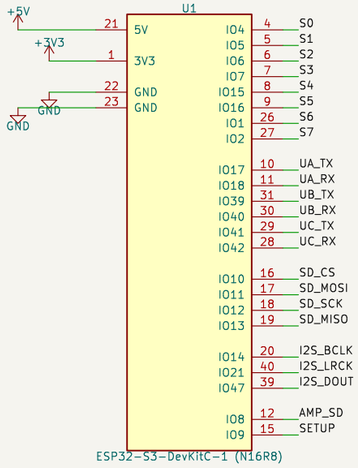
UART A = zum Empfaenger / vorigen Modul (Pin3 RX, Pin4 TX). UART B/C = weiter (Pin3 TX, Pin4 RX). Gesperrt am DevKit: IO0/3/45/46, IO19/20 (USB), IO26–37 (Flash/PSRAM), IO43/44 (UART0).

Lastversorgung 12–24V (VLOAD)

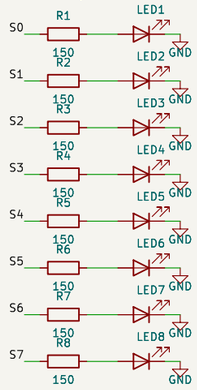


D1: TVS Anode GND, Kathode VLOAD

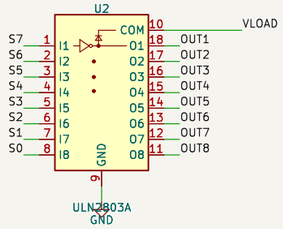
CN3 -> Versorgung TPA3116D2-Modul (+ / –)



Kontroll-LEDs (Sn -> R -> LED -> GND)

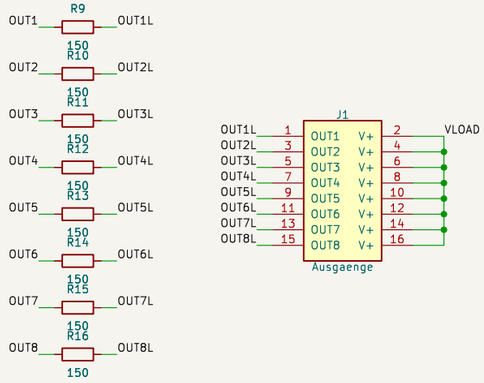


8-Kanal-Lastschalter (low-side), Eingaenge direkt vom ESP32

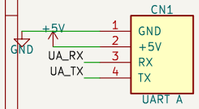


J1: ungerade Pins = OUT1..8 (ueber R9–R16), gerade Pins = VLOAD

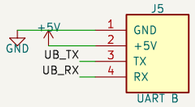
Ausgangswiderstaende (THT, je Verbraucher waehlen)



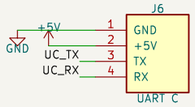
UART A (Empfaenger / vorheriges Modul)



UART B (naechstes Modul)



UART C (S.Port o.ae.)

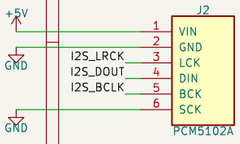


LED-Vorwiderstand R1–R8 bei 3,3V (ca. 5mA): Rot 270R, Gelb 270R, Gruen 220R, Blau 56R, Weiss 56R

AUDIO (nur eine Variante bestuecken)

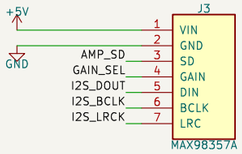
A) PCM5102A (6-pol.), SCK fest auf GND

Line-Out per Kabel zum TPA3116D2



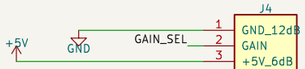
B) MAX98357A (7-pol.)

Lautsprecher direkt am Modul



J4: GND = 12dB, offen = 9dB,

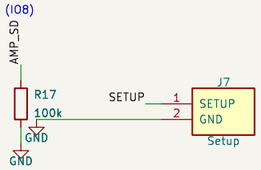
+5V = 6dB



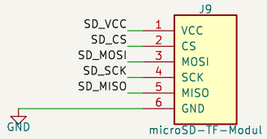
R17: Pull-down, stumm

beim Booten (IO8)

J7: Setup-Taster (IO9)

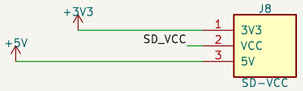


microSD TF-Card-Modul (6-pol., SPI auf IO10–13)



J8: Jumper 3V3 oder 5V je nach Modul

(Module mit Spannungsregler/Pegelwandler: 5V, reine 3V3-Module: 3V3)



Entwurf – vor Layout/Bestellung pruefen

Sheet: /
File: CRSF-Multiswitch-Sound-ESP32-S3.kicad_sch

Title: CRSF Multiswitch + Sound ESP32-S3 (ULN2803A, I2S, microSD)

Size: A3	Date: 2026-10-15	Rev: v0.1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1